Bridging the Gap: Statistical vs Machine Learning Methods

Performance, Cost, and Interpretability Trade-offs in the Age of Transparent AI

ML vs Statistical Models Research Team

Master Paris-Saclay
Applied Statistics

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Overview

1. Introduction
2. Methodology
3. Results
   - Regression Analysis
   - Classification Analysis
   - Time Series Analysis
   - Survival Analysis
4. The Trade-offs
   - Cost Analysis
   - Interpretability Analysis
5. Recommendations
6. Conclusions

Research Question

Core Question

How do Machine Learning and Statistical models compare across different problem domains in terms of:

- **Performance** (accuracy, R^2)
- **Cost** (training time, computational resources)
- **Interpretability** (explainability scores)

Connection to Interpretable ML Research

This work connects to the broader theme of **Transparent AI** and the need to bridge the gap between statistical interpretability and ML performance.

ML Models:

- Random Forest
- SVM
- MLP (Neural Network)

Statistical Models:

- OLS Regression
- GLM
- Logistic Regression

Scope of the Study

Category	Count	Range
Total Comparisons	62	Model-Dataset pairs
Problem Domains	4	Regression, Classification, Time Series, Survival
Datasets	13	150 to 35,000 samples
ML Models	38	Comparisons
Statistical Models	24	Comparisons

Connection to Current Research

This study contributes to the growing body of work on **Transparent AI** and interpretable machine learning, addressing the fundamental trade-offs between performance, cost, and interpretability highlighted in recent research (arXiv:2507.23535v1).

Problem Domains

- **Regression:** California Housing, Diabetes, Wine Quality, Energy Consumption

Evaluation Framework

Performance Metrics

- R^2 Score (Regression)
- Accuracy (Classification)
- RMSE, MAE, MAPE
- Concordance Index (Survival)

Efficiency Metrics

- Training Time
- Prediction Time
- Memory Usage

Interpretability Metrics

- Explainability Score (0-100)
- Feature Importance
- Statistical Significance (p-values)
- Confidence Intervals
- Model Diagnostics

Key analysis

Systematic comparison across multiple dimensions to identify trade-offs and best practices

Overall Performance Summary

ML Models

- Avg R^2: 0.289
- Avg Accuracy: 0.310
- Avg Training Time: 1.746s
- Avg Interpretability: 37.4/100

Statistical Models

- Avg R^2: 0.317
- Avg Accuracy: 0.297
- Avg Training Time: 0.061s
- Avg Interpretability: 88.3/100

Metric	ML	Statistical
Speed Advantage	1.0x	28.6x faster
Interpretability	37.4/100	88.3/100
Performance (R^2)	0.289	0.317

Winners by Category

Category	Winner	Performance
Best Overall Accuracy	Random Forest (ML)	$R^2 = 0.999$
Fastest Training	Linear Regression (ML)	0.0008s
Most Interpretable	OLS Regression (Stat)	100/100
Best for Production	GLM (Statistical)	Score = 0.9999
Most Efficient	GLM (Statistical)	0.002s avg

Key Insight

No single approach dominates all categories. The choice depends on the primary requirement.

Regression Results

Dataset	Best ML	R^2	Best Stat	R^2
California Housing	Random Forest	0.805	OLS	0.575
Diabetes	Linear Reg	0.453	OLS	0.453
Wine Quality	Linear Reg	0.787	OLS	0.787
Energy Consumption	Random Forest	0.970	GLM	0.811

Findings

- Random Forest excels on complex datasets (California Housing, Energy Consumption)
- Statistical models provide consistent baseline performance
- ML models show superior performance on high-dimensional datasets
- Statistical models train 32x faster on average

Classification Results

Dataset	Best ML	Acc.	Best Stat	Acc.
Iris	Random Forest	1.000	GLM	1.000
Breast Cancer	Random Forest	0.965	GLM	0.956
Wine Classification	SVM	0.982	GLM	0.956
Credit Default	Random Forest	0.999	Logistic Reg	0.999

ML Advantages:

- Higher accuracy on complex datasets
- Better handling of non-linear patterns
- Superior on high-dimensional data

Statistical Advantages:

- Full interpretability with p-values
- Confidence intervals for predictions
- Model diagnostics available

Time Series Results

Dataset	Best ML	R^2	Best Stat	R^2
Stock Prices	Random Forest	0.999	GLM	0.999
Temperature	MLP	0.965	GLM	0.972
Sales Forecasting	Random Forest	0.969	GLM	0.972

Key Observations

- Both approaches achieve exceptional performance ($R^2 > 0.99$) on Stock Prices
- Statistical models are significantly faster: GLM trains in 0.004s vs Random Forest's 0.699s
- Statistical models offer better interpretability for temporal relationships
- ML models capture complex non-linear temporal patterns

Survival Analysis Results

Dataset	Model	Concordance	Type
Customer Churn	Random Forest	0.000	ML
	Cox Regression	0.476	Statistical
Patient Survival	Random Forest	0.000	ML
	Cox Regression	0.536	Statistical

Domain-Specific Insight

Statistical models (Cox Regression, Kaplan-Meier) are specifically designed for survival analysis and provide:

- Domain-appropriate metrics (Concordance Index)
- Interpretable hazard ratios
- Statistical significance testing

The Fundamental Trade-offs

Three Critical Dimensions

Performance

- Accuracy
- R^2 Score
- Prediction Quality

Cost

- Training Time
- Computational Resources
- Infrastructure

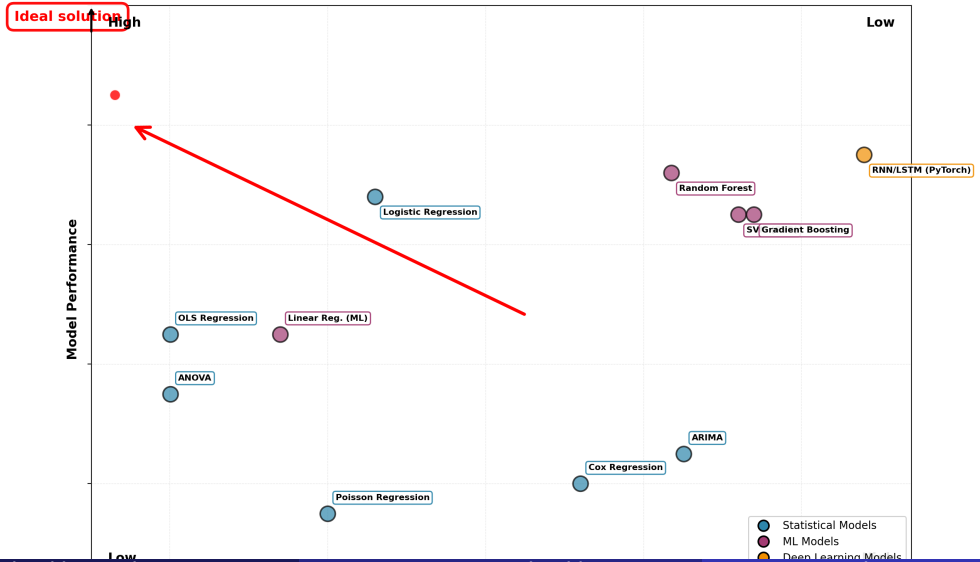
Interpretability

- Explainability
- Transparency
- Trust

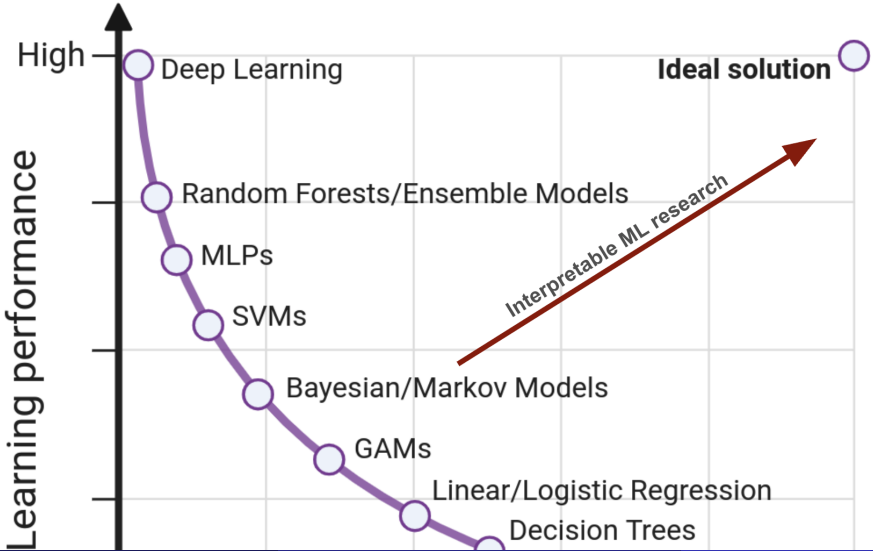
Key Insight

No single model excels in all three dimensions. The choice depends on your priorities.

Performance vs Training Cost Trade-off



Performance vs Interpretability Trade-off



Performance vs Interpretability Trade-off (Analysis)

High Performance, Low Interpretability

- Deep Learning
- Random Forests
- MLPs
- SVMs

High Interpretability, Moderate Performance

- Linear/Logistic Regression
- Decision Trees
- GAMs
- Rule-based Learning

The Ideal Solution

- High Performance
- High Interpretability
- Currently: Research frontier
- Future: Interpretable ML

Research Direction

Interpretable ML research aims to move models from the curve toward the ideal solution, combining best of both worlds.

The Three-Dimensional Trade-off

Performance × Cost × Interpretability

Statistical Models

- ✓ High Interpretability
- ✓ Low Cost
- × Moderate Performance
- **Best for:** Research, Compliance, Speed

ML/DL Models

- ✓ High Performance
- × Higher Cost
- × Low Interpretability
- **Best for:** Production, Complex Patterns

The Challenge

How do we achieve High Performance + Low Cost + High Interpretability?

Speed Analysis

Ranking	Model	Training Time
1	Linear Regression (ML)	0.0008s
2	GLM (Statistical)	0.002s
3	OLS Regression	0.003s
...	...	...
Last	Random Forest	9.644s

Statistical Models:

- 32x faster on average
- Best for real-time applications
- Low computational cost

ML Models:

- Longer training times
- Higher computational cost
- Better for batch processing

Interpretability Gap

Interpretability Scores (0-100)

Top Interpretable Models

1. OLS Regression: 100/100
2. Logistic Regression: 90/100
3. GLM: 90/100

Low Interpretable Models

1. SVM: 30/100
2. MLP: 30/100
3. Linear Reg (ML): 30/100
4. Random Forest: 50/100

Statistical Models Provide

- Coefficients with p-values and confidence intervals
- Model diagnostics (AIC, BIC, F-statistics)
- Statistical significance tests
- Residual analysis

When to Choose Statistical Models

Choose Statistical Models When:

- **Interpretability is critical** - Need p-values and confidence intervals
- **Regulatory compliance** - Transparent decision-making required
- **Hypothesis testing** - Understanding causal relationships
- **Small to medium datasets** - Optimal performance on limited data
- **Real-time applications** - Fast training and prediction needed

Best Models: OLS Regression, GLM, Logistic Regression, Cox Regression

When to Choose ML Models

Choose ML Models When:

- **Maximum accuracy is priority** - Need best predictive performance
- **Large, complex datasets** - High-dimensional or non-linear patterns
- **Production systems** - Computational resources available
- **Pattern discovery** - Exploratory analysis on complex data
- **Non-linear relationships** - When linear assumptions fail

Best Models: Random Forest, SVM, MLP (Neural Networks)

Use Case Recommendations

Use Case	Recommended	Rationale
Research & Development	Statistical	Interpretability, hypothesis testing
Production Systems	ML	Maximum accuracy, scalability
Regulatory Compliance	Statistical	Transparency, auditability
Real-time Applications	Statistical	Fast training/prediction
Large-scale Data	ML	Better scalability, complex patterns

Hybrid Approach

Consider combining both methodologies:

- Use statistical models for baseline and interpretability
- Use ML models for final production deployment
- Compare results from both approaches for validation

Key Takeaways

1. **No universal winner** - Both approaches have distinct advantages
2. **Statistical models excel in:**
   - Interpretability (88.3/100 vs 37.4/100)
   - Speed (28.6x faster training)
   - Regulatory compliance
3. **ML models excel in:**
   - Complex dataset performance
   - Scalability to large data
   - Capturing non-linear patterns
4. **The optimal choice depends on:**
   - Primary requirement (accuracy vs interpretability)
   - Dataset characteristics
   - Computational constraints
   - Regulatory requirements

Strategic Recommendations

For Organizations

- **Develop expertise in both** ML and statistical modeling
- **Establish clear guidelines** for model selection based on requirements
- **Invest in tools** that support both approaches
- **Consider hybrid systems** combining both methodologies

For Practitioners

- Start with statistical models for baseline and interpretability
- Use ML models when accuracy is critical and data is complex
- Always validate ML results with statistical diagnostics when possible
- Document trade-offs in model selection decisions

Bridging the Gap: Interpretable ML Research

The Vision

Achieve the ideal solution: High Performance + High Interpretability + Low Cost

Current Approaches:

- **Model-Agnostic Methods**
  - SHAP values
  - LIME explanations
  - Partial dependence plots
- **Inherently Interpretable Models**
  - Generalized Additive Models (GAMs)
  - Explainable Boosting Machines
  - Neural Additive Models

Research Directions:

- Hybrid Statistical-ML frameworks
- Post-hoc interpretability techniques
- Transparent AI architectures
- Regulatory-compliant ML systems
- Explainable deep learning

The Path Forward

Moving from the trade-off curve toward the ideal solution requires interdisciplinary research combining statistics, ML, and human-centered AI.

Future Research Directions

1. **Hybrid Models** - Combining ML performance with statistical interpretability
2. **Interpretable ML** - Advanced techniques for making ML models more transparent (as shown in bridging_gap.png)
3. **Cost-Efficient ML** - Optimizing training costs while maintaining performance (as shown in CostVsPerformance.png)
4. **Automated Model Selection** - Systems that automatically choose between approaches based on trade-offs
5. **Transparent AI Frameworks** - Regulatory-compliant ML systems with full interpretability
6. **Domain-Specific Analysis** - Deeper analysis within specific industries (finance, healthcare, etc.)

Thank You!

Questions?

Contact: ML vs Statistical Models Research Team
Institution: Master Paris-Saclay, Applied Statistics
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References & Resources

- Rudin, C., & Radin, J. (2019). Why Are We Using Black Box Models in AI When We Don't Need To? A Lesson From An Explainable AI Competition. Harvard Data Science Review.
- Arrieta, A. B., et al. (2020). Explainable Artificial Intelligence (XAI): Concepts, Taxonomies, Opportunities and Challenges. Information Fusion, 58, 82-115.
- Hastie, T., Tibshirani, R., & Friedman, J. (2009). The Elements of Statistical Learning. Springer.
- James, G., Witten, D., Hastie, T., & Tibshirani, R. (2013). An Introduction to Statistical Learning. Springer.
- Murphy, K. P. (2012). Machine Learning: A Probabilistic Perspective. MIT Press.
- Cox, D. R. (1972). Regression Models and Life-Tables. Journal of the Royal Statistical Society.
- Kaplan, E. L., & Meier, P. (1958). Nonparametric Estimation from Incomplete Observations. JASA.
- **Paper:** Transparent AI: The Case for Interpretability and Explainability (arXiv:2507.23535v1)

Data & Code

- Comprehensive results: `comprehensive_results/`
- Visualizations: `imgs/`
- Documentation: `docs/`
- Analysis scripts: `notebooks/`